

# Canonical-height distributions for constant-coefficient Hénon maps

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## Abstract

We determine the forward-plus-backward canonical-height distribution of every polynomial point for every constant-coefficient Hénon map  $H(x, y) = (y, f(y) - ax)$  over  $\mathbb{F}_q$ , with  $\deg f = d \geq 2$  and  $a \neq 0$ . The distribution depends only on  $q$  and  $d$ , including in inseparable cases. A unique degree valley in each nonconstant orbit gives two disjoint classes of representatives with exact multiplicities. Summing their integer time shifts yields an explicit height Dirichlet series, absolutely convergent exactly for  $\Re s > 1$  and meromorphic for  $\Re s > 0$ . We determine every pole after combining the sectors: all are simple except for a specified lattice of double poles on  $\Re s = 1$ . The surviving pole lattices accumulate at every point of the imaginary axis, which is a natural boundary even for meromorphic continuation. Direct counting gives  $\#\{P : \widehat{h}(P) \leq B\} = \frac{d-1}{d+1}(q-1)q^{\lfloor B \rfloor + 1}B + O_{q,d}(q^B)$  for arbitrary real  $B$ . The fractional-part oscillation is retained in the leading term. The domain is  $\mathbb{F}_q[t]^2$  and the series counts heights, not periodic orbits.

## 1 Introduction

For a fixed nonperiodic orbit of a Hénon map, forward and backward canonical heights have opposite geometric scaling. Counting all polynomial points requires an additional step: the different orbits must be parametrized without overlap, with the correct arithmetic multiplicity. For constant coefficients over a finite field, this step has an exact answer. Each nonconstant orbit has a unique degree valley of one of two types. Their multiplicities do not depend on the coefficients of the map.

We use this classification to determine the full two-sided canonical-height distribution on  $\mathbb{F}_q[t]^2$ . Its analytic structure is not captured by a single leading pole. The sectors coming from neighboring time shifts meet at the same candidate poles, and their residues must be combined before any natural-boundary conclusion is justified. We compute the combined residue explicitly and prove that it never vanishes at the proposed simple poles.

The resulting contribution has three parts.

1. An exhaustive, disjoint edge/turn parametrization gives exact height multiplicities for every prime power, every degree at least two and every allowed constant coefficient.
2. The associated series has absolute-convergence abscissa one, an explicit complete simple/double pole divisor in  $\Re s > 0$ , and a meromorphic natural boundary at  $\Re s = 0$ .
3. A direct count gives the real-height asymptotic with its nonconstant fractional-part oscillation. Replacing the floor in the exponential main term by a real exponent would change the result.

**Classical height inputs.** Kawaguchi constructs local Green functions and global canonical heights for regular polynomial automorphisms [3]. The height limits, functional equations and two-sided orbit scaling are classical inputs. The global results there are formulated over number fields, and the bounded-height orbit count in Corollary D concerns a given orbit and naive height. It does not give the distribution over all polynomial points studied here. Our required specialization is proved directly by degrees.

Ingram develops canonical heights and non-Archimedean escape for Hénon maps [2]. The introduction and Section 2 of arXiv:1111.3609v1, in particular Lemma 2.1, supply the nearby framework used for comparison. The direct degree argument below handles arbitrary nonzero leading coefficient without a separability assumption. We do not claim a new canonical-height construction or a general Northcott theorem. The bibliography uses the verified journal issue metadata, while the text comparison remains to the accessed arXiv version.

**Height-zeta counting.** Hsia's published development of dynamical height zeta functions [1] is a predecessor of the one-variable counting program. We use its bibliographic attribution, not an unaccessed theorem: the original full article was not available for the present comparison. Takehira's account attributes the definition, via Hsia, to Silverman's unpublished 1994 seminar talk; that talk was also not accessed. Neither is a proof dependency here.

Takehira gives explicit one-variable function-field height-zeta formulas and treats the effect of several vertical-line poles on height counting [4]. Our comparison is to the introduction, Section 2.2, Condition 3.1, Theorem 3.4 and the opening of Section 4 in arXiv:2404.00955v1. Its finite local height-discrepancy partition does not apply to the present whole two-dimensional domain: for an edge with both coordinate degrees  $m$ , the two-sided height is  $2m$  and the naive height is  $m$ . The discrepancy is therefore unbounded even with constant coefficients. The theorem below retains the oscillation explicitly, without a single-pole Tauberian assumption. These source comparisons are bounded by the stated text access and do not assert worldwide priority.

Section 2 fixes the domain and states the distribution and analytic conclusions. Sections 3 and 4 prove the degree-valley census and sum the sectors. Section 5 combines their residues and proves the boundary; Section 6 counts points directly.

## 2 Heights, the exact distribution and the main theorems

Fix a prime power  $q$ , an integer  $d \geq 2$ , a polynomial  $f \in \mathbb{F}_q[X]$  of degree exactly  $d$ , and  $a \in \mathbb{F}_q^*$ . Let  $R = \mathbb{F}_q[t]$  and

$$H(x, y) = (y, f(y) - ax), \quad H^{-1}(x, y) = (a^{-1}(f(x) - y), x). \quad (2.1)$$

For  $P = (x, y) \in R^2$  put  $h(P) = \max(0, \deg x, \deg y)$ , with  $\deg 0 = -\infty$ . Our canonical heights are in units of  $\log q$ :

$$\hat{h}^\pm(P) = \lim_{n \rightarrow \infty} d^{-n} h(H^{\pm n} P), \quad \hat{h}(P) = \hat{h}^+(P) + \hat{h}^-(P). \quad (2.2)$$

Their existence follows from the degree patterns proved below. Define the height Dirichlet series and its counting function by

$$Z_H(s) = \sum_{P \in R^2} q^{-s\hat{h}(P)}, \quad N_H(B) = \#\{P \in R^2 : \hat{h}(P) \leq B\}. \quad (2.3)$$

Here  $q^z = \exp(z \log q)$  with the positive real logarithm of  $q$ .

For  $M \geq 1$ , define the integer weight

$$D_M = \begin{cases} (q-1)^2 q^{(d+1)r}, & M = dr, \quad r \geq 1, \\ (q-1)q^{(d+1)r+j+1}, & M = dr + j, \quad r \geq 0, \quad 1 \leq j < d. \end{cases} \quad (2.4)$$

**Theorem 2.1** (Coefficient-uniform distribution). *The limits (2.2) exist for every polynomial point and  $h(P) \leq \widehat{h}(P)$ . The full distribution of  $\widehat{h}$  depends only on  $q, d$ , not on the leading coefficient of  $f$ , its lower coefficients, or  $a$ . Its exact series is*

$$\begin{aligned} Z_H(s) &= q^2 + (q-1)^2 \sum_{\substack{m, n \geq 1 \\ m < dn, \quad n < dm}} q^{m+n} \sum_{k \in \mathbb{Z}} q^{-s(d^{-k}m + d^k n)} \\ &\quad + \sum_{M \geq 1} D_M \sum_{k \in \mathbb{Z}} q^{-sM(d^{-k} + d^{k-1})}. \end{aligned} \quad (2.5)$$

*This is first an identity of discrete nonnegative counting measures; it is an identity of analytic series wherever absolutely convergent. Height zero has multiplicity exactly  $q^2$ , and every bounded-height set is finite.*

**Theorem 2.2** (Complete poles and meromorphic boundary). *The series  $Z_H$  converges absolutely exactly when  $\Re s > 1$ , and it has a meromorphic continuation to  $\Re s > 0$ . Put*

$$L_k = d^k + d^{1-k}, \quad s_{k,\ell} = \frac{d+1}{L_k} + \frac{2\pi i \ell}{L_k \log q}, \quad k \geq 1, \quad \ell \in \mathbb{Z}. \quad (2.6)$$

*These are exactly its poles in  $\Re s > 0$ . The poles with  $k = 1$  and  $(d+1) \mid \ell$  are double; all others are simple. At every double pole, the coefficient of  $(s - s_{1,\ell})^{-2}$  is*

$$\frac{(q-1)^2(d-1)}{(d+1)(\log q)^2}. \quad (2.7)$$

*An explicit nonzero residue at every simple pole is given in (5.5). Every point of the imaginary axis is an accumulation point of poles. The axis is consequently a natural boundary for meromorphic continuation.*

**Theorem 2.3** (Real-height counting). *As  $B \rightarrow \infty$  through arbitrary real values,*

$$N_H(B) = \frac{d-1}{d+1} (q-1) q^{\lfloor B \rfloor + 1} B + O_{q,d}(q^B). \quad (2.8)$$

*In particular, writing  $\{B\} = B - \lfloor B \rfloor$ ,*

$$\frac{N_H(B)}{Bq^B} = \frac{d-1}{d+1} (q-1) q^{1-\{B\}} + O_{q,d}(B^{-1}).$$

*The leading coefficient is not constant along real height bounds.*

All three statements concern the entire polynomial-point domain, not all of  $\mathbb{F}_q(t)^2$ . Constants form a finite invariant set; every nonconstant polynomial orbit will be shown to escape in both directions. Thus (2.3) is not an ordinary periodic-orbit zeta function.

### 3 Unique degree valleys and their multiplicities

Write the entire orbit as  $H^j P = (x_j, x_{j+1})$  and set  $b_j = \max(0, \deg x_j)$ . Then

$$x_{j+1} + ax_{j-1} = f(x_j), \quad j \in \mathbb{Z}. \quad (3.1)$$

If  $b_j \geq 1$  and  $b_j \geq b_{j+1}$ , the right side has degree  $db_j > b_{j+1}$ , so  $b_{j-1} = db_j$ . Repetition forces geometric growth in the backward direction. If  $b_j \leq b_{j+1}$  and  $b_{j+1} \geq 1$ , the same comparison gives  $b_{j+2} = db_{j+1}$  and geometric forward growth. Two consecutive zeros would put the point in  $\mathbb{F}_q^2$ , which is invariant under both maps in (2.1); the entire orbit would consist of constant points.

Every nonconstant orbit therefore escapes in both directions. Indeed, if one direction has not yet been covered by a nondecreasing adjacent pair in that orientation, failure to find such a pair would give an infinite strictly decreasing sequence of nonnegative integers. A positive plateau instead forces escape, and a zero plateau implies constant coordinates. The global minimum of  $(b_j)$  is attained. Both-sided escape also excludes periodicity, so all integer iterates on a nonconstant orbit are distinct.

**Proposition 3.1** (Disjoint orbit representatives). *Every nonconstant orbit contains exactly one representative of one of the following two types, and never representatives of both types.*

1. An edge:  $(x, y)$  with  $m = \deg x \geq 1$ ,  $n = \deg y \geq 1$ ,  $m < dn$  and  $n < dm$ .
2. A turn:  $(x, y)$  with  $M = \deg x \geq 1$ ,  $d \max(0, \deg y) \leq M$  and  $\deg(f(y) - ax) = M$ .

*Proof.* At a strict minimum  $b_j = n$ , let the neighboring degrees be  $b, c > n$ . For  $n \geq 1$  and  $b \neq c$ , the recurrence forces  $\max(b, c) = dn$ . The pair formed from  $x_j$  and its smaller neighbor has positive degrees satisfying both strict edge inequalities. If  $b = c = M$ , cancellation in the recurrence forces  $M \geq dn$  for  $n \geq 1$ . For  $n = 0$  the same inequality holds and the equal-neighbor case is forced, since unequal positive neighbors could not have a constant sum. The ordered pair  $(x_{j-1}, x_j)$  is then a turn. If a minimum occurs at two consecutive equal degrees, those degrees are positive and the adjacent pair is an edge. Three consecutive equal positive degrees contradict the escape comparison. These cases exhaust the minima and establish existence.

For uniqueness, an edge of degrees  $(m, n)$  has exterior degrees  $dm$  on the left and  $dn$  on the right, and all farther degrees grow by  $d$ . No other adjacent pair satisfies the edge inequalities, and no turn occurs. A turn has the pattern

$$\dots, d^2 M, dM, M, n, M, dM, d^2 M, \dots, \quad 0 \leq n \leq M/d. \quad (3.2)$$

The falling-to-rising orientation selects a unique ordered pair. The reversed adjacent pair is not another turn, and the pattern has no edge. This proves disjointness and uniqueness.  $\square$

**Lemma 3.2** (Heights at all shifts). *For an edge representative of degrees  $(m, n)$ ,*

$$(\hat{h}^-(P), \hat{h}^+(P)) = (m, n), \quad \hat{h}(H^k P) = d^{-k}m + d^k n. \quad (3.3)$$

*For a turn representative of degree  $M$ ,*

$$(\hat{h}^-(P), \hat{h}^+(P)) = (M, M/d), \quad \hat{h}(H^k P) = M(d^{-k} + d^{k-1}). \quad (3.4)$$

*Constants have height zero. For every polynomial point,  $h(P) \leq \hat{h}(P)$ . An edge uniquely minimizes its total height on its orbit; a turn has exactly two adjacent minima, at  $k = 0, 1$ .*

*Proof.* The geometric degree patterns prove both limits directly. For a turn, forward geometric growth begins one step later, accounting for the factor  $1/d$ . Shifting either limit gives the displayed formulas for every  $k \in \mathbb{Z}$ . In those patterns, the largest coordinate degree away from the center is one of the two positive height summands. At the center it is at most their sum, proving  $h \leq \hat{h}$ . This also gives finiteness of bounded-height sets, contained in the degree box  $h(P) \leq \lfloor B \rfloor$ .

For an edge,  $1/d < n/m < d$ . The first height difference in each direction from  $k = 0$  is strictly positive, and subsequent differences increase. For a turn the formula is invariant under  $k \mapsto 1 - k$  and increases away from  $k = 0, 1$ .  $\square$

There are  $(q - 1)q^m$  polynomials of degree  $m \geq 1$  and  $q$  constants. Every pair in the open edge cone is admissible, so its exact multiplicity is  $(q - 1)^2 q^{m+n}$ . To count turns of degree  $M$ , first impose  $d \deg y < M$ . The second turn condition is automatic and the number of pairs is

$$(q - 1)q^M q^{\lfloor (M-1)/d \rfloor + 1}. \quad (3.5)$$

If  $M = dr$  and  $\deg y = r \geq 1$ , exactly one nonzero leading coefficient of  $x$  would cancel that of  $f(y)$ . For each  $y$  there are therefore  $(q - 2)q^M$  choices of  $x$ , giving  $(q - 1)(q - 2)q^{(d+1)r}$  boundary pairs. At  $q = 2$  this boundary portion is zero; the strict portion is still present. Adding the two disjoint portions proves (2.4).

| Representative | Multiplicity        | Height at integer shift $k$ |
|----------------|---------------------|-----------------------------|
| Edge $(m, n)$  | $(q - 1)^2 q^{m+n}$ | $d^{-k}m + d^k n$           |
| Turn $M$       | $D_M$ of (2.4)      | $M(d^{-k} + d^{k-1})$       |
| Constant point | $q^2$ in total      | 0                           |

Table 1: A disjoint census of all polynomial points. Edge and turn representatives satisfy Proposition 3.1; each of their integer shifts is a distinct point. The two equal minimum heights of a turn are two points, not two representatives.

Only degrees and nonvanishing of the specified coefficients entered these counts. They hold over every prime power, including when  $f$  is inseparable. Summing all shifts and the  $q^2$  constant points proves Theorem 2.1.

## 4 Sector formulas and meromorphic convergence

The positive census (2.5) becomes a normally convergent sum of rational sectors after two elementary summations. First define the open-cone series

$$T_d(A, B) = \sum_{\substack{m, n \geq 1 \\ m < dn, n < dm}} A^m B^n. \quad (4.1)$$

Subtracting the two disjoint closed tails from the positive quadrant gives, first on its domain of absolute convergence,

$$T_d(A, B) = \frac{AB}{(1 - A)(1 - B)} - \frac{A^d B}{(1 - A)(1 - A^d B)} - \frac{AB^d}{(1 - B)(1 - AB^d)}. \quad (4.2)$$

The tails are  $m \geq dn$  and  $n \geq dm$ ; they cannot intersect for positive  $m, n$  and  $d \geq 2$ .

The factors  $1 - A$  and  $1 - B$  in this expression do not by themselves give cone poles. To make the cancellation explicit, let  $\mathcal{P}_d$  be the finite set of lattice points in

$$\{\alpha(d, 1) + \beta(1, d) : 0 < \alpha \leq 1, 0 < \beta \leq 1\}.$$

A point in the strict cone has unique positive real coordinates along the independent vectors  $(d, 1), (1, d)$ . Subtract the ceilings of these coordinates minus one. The remainder is still a lattice point and lies in  $\mathcal{P}_d$ , while the subtracted coefficients are nonnegative integers. This representation is unique. Hence

$$T_d(A, B) = \frac{\sum_{(r,t) \in \mathcal{P}_d} A^r B^t}{(1 - A^d B)(1 - AB^d)}. \quad (4.3)$$

Both coordinates of every point of  $\mathcal{P}_d$  are positive. Formula (4.3) proves that the apparent axial singularities in (4.2) are removable away from the two actual cone denominators.

Second, summing the residue classes of  $M$  in (2.4) gives

$$D_{q,d}(z) = \sum_{M \geq 1} D_M z^M = \frac{(q-1)^2 q^{d+1} z^d + (q-1)q \sum_{r=1}^{d-1} (qz)^r}{1 - q^{d+1} z^d}. \quad (4.4)$$

For  $k \in \mathbb{Z}$  put

$$\begin{aligned} A_k &= q^{1-sd^{-k}}, & B_k &= q^{1-sd^k}, & C_k(s) &= (q-1)^2 T_d(A_k, B_k), \\ z_k &= q^{-s(d^{k-1}+d^{-k})}, & E_k(s) &= D_{q,d}(z_k). \end{aligned} \quad (4.5)$$

The symmetries  $C_{-k} = C_k$  and  $E_{1-k} = E_k$  turn (2.5) into

$$Z_H(s) = q^2 + C_0(s) + 2 \sum_{k \geq 1} C_k(s) + 2 \sum_{k \geq 1} E_k(s). \quad (4.6)$$

**Proposition 4.1** (Convergence and candidate poles). *The positive series converges absolutely exactly for  $\Re s > 1$ . Formula (4.6) continues it meromorphically to  $\Re s > 0$ , with no possible poles except (2.6).*

*Proof.* With  $L_0 = L_1 = d + 1$ , the denominators in  $C_k$  are

$$1 - q^{d+1-sL_k}, \quad 1 - q^{d+1-sL_{k+1}},$$

and the denominator of  $E_k$  is the first of these. The sequence  $L_k$  strictly increases for integer  $k \geq 1$ . Thus all sector power series are absolutely convergent when  $\Re s > 1$ .

On a compact subset of  $\Re s \geq \varepsilon > 0$ , as  $k \rightarrow \infty$  the quantities  $A_k$  are bounded,  $B_k = O(q^{-\varepsilon d^k})$  and  $z_k = O(q^{-\varepsilon d^{k-1}})$ . The numerator of (4.3) has positive  $B$  exponents, while both denominators tend uniformly to one. Its tail and the tail of (4.4) are consequently uniformly summable. This proves absolute convergence of the sector sum in  $\Re s > 1$  and locally normal convergence of its holomorphic tail in  $\Re s > 0$ . Only finitely many meromorphic sectors remain on any compact subset, giving the stated continuation and the complete list of candidate poles.

For the converse on absolute convergence, put  $j = m + n$ . The number of positive pairs of sum  $j$  in the strict cone is

$$b_j = j - 2\lfloor j/(d+1) \rfloor - 1 = \frac{d-1}{d+1}j + O_d(1), \quad j \geq 1. \quad (4.7)$$

Indeed the strict bounds on  $m$  are  $j/(d+1) < m < dj/(d+1)$ . The central edge terms alone are  $(q-1)^2 \sum_{j \geq 1} b_j q^{(1-s)j}$ , which diverge for real  $s \leq 1$ . At complex  $s$ , absolute values replace  $s$  by  $\Re s$ . This proves the exact abscissa.  $\square$

## 5 Combined residues and the natural boundary

Candidate poles of separate sectors need not survive their sum. We combine all sectors at each lattice point before deciding its order. Fix  $k \geq 2$  and  $s_0 = s_{k,\ell}$ . Since the  $L_k$  are strictly increasing for  $k \geq 1$ , only  $C_{k-1}$ ,  $C_k$  and  $E_k$  can have a pole there, each with weight two in (4.6). Formula (4.2) gives the coefficients of  $(1 - q^{d+1-sL_k})^{-1}$  from the two cone sectors, evaluated at  $s_0$ , as

$$\frac{(q-1)^2}{B_{k-1}(s_0)-1}, \quad \frac{(q-1)^2}{A_k(s_0)-1}.$$

Set  $A = A_k(s_0)$  and  $v = (qz_k(s_0))^{-1}$ . The pole condition implies

$$v^d = q, \quad AB_{k-1}(s_0) = q^2 z_k(s_0) = q/v. \quad (5.1)$$

Adding the turn numerator in (4.4), the full coefficient before the common weight two is

$$R = (q-1)^2 \left( \frac{1}{A-1} + \frac{A}{q/v-A} + 1 \right) + (q-1)q \sum_{r=1}^{d-1} v^{-r}. \quad (5.2)$$

Using  $v^d = q$ , the finite geometric sum is  $(q-v)/(q(v-1))$ . Also

$$\frac{1}{A-1} + \frac{A}{q/v-A} + 1 = \frac{A(q-v)}{(A-1)(q-vA)}.$$

The remaining numerator identity is

$$(q-1)A(v-1) + (A-1)(q-vA) = (q-A)(vA-1).$$

Consequently the exact aggregate factorizes as

$$R = \frac{(q-1)(q-v)(q-A)(vA-1)}{(v-1)(A-1)(q-vA)}. \quad (5.3)$$

Let  $u = q^{1/d}$ . At the selected lattice point  $|v| = u$ , and for  $k \geq 2$ ,

$$q/u < |A| = q^{1-(d+1)/(d^k L_k)} < q. \quad (5.4)$$

All numerator and denominator factors in (5.3) are nonzero:  $1 < |v| < q$ ,  $1 < |A| < q$ , and  $|vA| > q > 1$ . Thus every candidate at level  $k \geq 2$  is a genuine simple pole. Since the derivative of  $1 - q^{d+1-sL_k}$  at  $s_0$  is  $L_k \log q$ , its residue is

$$\text{Res}_{s=s_0} Z_H(s) = \frac{2R}{L_k \log q}, \quad R \text{ given by (5.3)}. \quad (5.5)$$

**The level-one collision.** For  $k = 1$ , both cone denominators of  $C_0$  coincide. Put  $w = B_0(s_0) = q^{1-s_0}$ . When  $w \neq 1$ , formula (4.2) gives singular coefficient  $2(q-1)^2/(w-1)$  from  $C_0$ . Although  $C_0$  has weight one in the sum, the coincident cone factors have thus supplied the same weight two needed in (5.2). The other sectors  $C_1, E_1$  already have weight two. Equations (5.3) and (5.5) hold unchanged.

At this level  $|A| = q/u > 1$  and  $|vA| = q$ . The factor  $q - vA$  vanishes exactly when  $w = 1$ , by (5.1). Excluding that case, it is nonzero, and every numerator factor is still nonzero by its modulus. Thus these level-one points are genuine simple poles.

Finally  $w = 1$  exactly when  $(d+1) \mid \ell$ . From (4.7), write  $b_j = \alpha j + \rho_j$ , where  $\alpha = (d-1)/(d+1)$  and  $\rho_j$  is bounded and periodic of period  $d+1$ . The series of  $\rho_j w^j$  is rational with denominator  $1 - w^{d+1}$  and hence has at most a simple pole at  $w = 1$ . The linear part gives  $\alpha w/(1-w)^2$ . Since  $w = q^{1-s}$  and  $w(s_0) = 1$ , the coefficient of  $(s - s_0)^{-2}$  in  $C_0$  is precisely (2.7). It is positive. The other sectors have at most simple poles, so cannot cancel it. We have now proved the entire pole divisor, including every root-of-unity phase and the cases  $q = 2$  or  $d = 2$ .

**Accumulation at every boundary point.** As  $k \rightarrow \infty$ ,  $(d+1)/L_k \rightarrow 0$  and  $2\pi/(L_k \log q) \rightarrow 0$ . Given  $\tau \in \mathbb{R}$ , choose  $\ell_k$  to be an integer nearest to  $\tau L_k \log q/(2\pi)$ . The genuine poles  $s_{k, \ell_k}$  converge to  $i\tau$ . A meromorphic continuation across that point would have poles accumulating at an interior point, which is impossible for a meromorphic function agreeing with the initial finite-valued series. Hence no point of the axis admits such a continuation. This completes the proof of Theorem 2.2.

## 6 Counting at arbitrary real height bounds

We prove Theorem 2.3 directly from the nonnegative census, so the multiple poles on  $\Re s = 1$  do not create an unjustified single-pole asymptotic. Set  $N = \lfloor B \rfloor$  and  $\alpha = (d-1)/(d+1)$ . The central edge contribution, corresponding to  $k = 0$ , is

$$(q-1)^2 \sum_{j \leq N} b_j q^j = \alpha(q-1)q^{N+1}N + O_{q,d}(q^N). \quad (6.1)$$

Indeed (4.7) has a bounded error, and differentiation of the finite geometric sum gives

$$\sum_{j \leq N} j q^j = \frac{q^{N+1}N}{q-1} + O_q(q^N).$$

The adjacent edge shifts  $k = 1, -1$  each contribute  $O_{q,d}(q^B)$ . For  $k = 1$  set  $e = dn - m \geq 1$ . Its height and multiplicity exponent become

$$d^{-1}m + dn = (d+1)n - e/d, \quad m + n = (d+1)n - e.$$

For fixed  $e$ , enlarge the degree range to all positive  $n$  satisfying the height bound. The geometric sum of weights is at most a constant depending on  $q, d$  times  $q^{B-(1-1/d)e}$ . Summing over  $e \geq 1$  converges. Interchanging  $m, n$  treats the negative shift. The two central turn shifts  $k = 0, 1$  also contribute  $O_{q,d}(q^B)$ , since

$$D_M \leq c_{q,d} q^{(1+1/d)M}$$

and their height is  $(1+1/d)M$ ; the remaining sum in  $M$  is geometric.

All other shifts have a strict exponential gap. Put

$$\alpha_2 = \frac{d+1}{d^2 + d^{-1}} < 1. \quad (6.2)$$

For edge shifts  $|k| \geq 2$ , the cone inequalities give

$$m + n \leq \alpha_2(d^{-k}m + d^k n). \quad (6.3)$$

For  $k > 0$ , the ratio of the left side to the height increases with  $m/n$  on  $1/d < m/n < d$ . Its supremum is  $(d+1)/L_k \leq \alpha_2$ , proving the claim; negative indices follow by symmetry. For turns with  $k \notin \{0, 1\}$ , the analogous inequality is

$$(1+1/d)M \leq \alpha_2 M(d^{-k} + d^{k-1}). \quad (6.4)$$



The first exterior turn shifts give equality in the ratio bound, and the denominator grows farther away from the center.

At height at most  $B$ , the representative degrees are at most  $B$  by Lemma 3.2. There are only  $O_d(1 + \log(1 + B))$  admissible integer shifts: every edge height is at least  $d^{|k|}$ , and every turn height is at least  $d^{\max(k-1, -k)}$ . Thus all the exterior contributions are

$$O_{q,d}\left(B^2(1 + \log(1 + B))q^{\alpha_2 B}\right) = O_{q,d}(q^B).$$

The strict gap  $1 - \alpha_2$  absorbs the polynomial and logarithmic factors. The  $q^2$  constant points are harmless. Combining these estimates with (6.1), then replacing  $N$  by  $B$  only in its linear factor, proves (2.8). The latter replacement costs  $O_{q,d}(q^B)$ , but replacing  $\lfloor B \rfloor$  in the exponent would change the leading oscillation. This completes the proof for arbitrary real  $B$ .

## 7 Scope and reproducibility

The exact distribution follows from an all-orbit degree-valley census, not from the canonical-height definition alone. Its coefficient uniformity includes arbitrary nonzero constant leading coefficients, all lower coefficients, every nonzero determinant and inseparable polynomials. The natural boundary follows only after the neighboring sectors' residues have been combined and shown nonzero. The point-count asymptotic retains the lattice dependence on a real height bound.

The accompanying source package includes the exact diagnostic `check_height_valleys.py` and its frozen execution receipt. It compared literal polynomial iteration with the independently specified valley weights in eleven prime-field cases, for 77,974 polynomial pairs. Heights were exact rational numbers, with escape certified by strict degree domination; all tested distributions agreed. The test varies coefficients, degree and determinant but does not establish the prime-power or infinite-parameter theorem. Those quantifiers, the combined residue and the analytic continuation are proved in the article. The script was not rerun merely to produce this manuscript.

Our domain restriction is substantive. The result does not count all rational-function points, treat nonconstant coefficients, or assert a height distribution over an arbitrary constant field. In particular, it does not classify the nonconstant-parameter periodic loci of a different Hénon family. Source version and access limits are retained in the accompanying citation audit; an independent internal review is not human journal peer review or a worldwide novelty certificate.

Finally,  $Z_H$  is a height Dirichlet series. Its pole divisor and mean counting exponent are not a target zero-counting correspondence. No target Euler factors, root numbers, automorphy, arithmetic divisor matching or Hilbert–Pólya realization follow from these source results.

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